
🟢 BASIC LEVEL ANSWERS

1. What is NumPy?

NumPy is a Python library used for fast numerical computations and working with arrays.

2. Why NumPy instead of lists?

NumPy is faster, uses less memory, and supports mathematical operations directly.

3. What is ndarray?

ndarray is NumPy's main array object (n-dimensional array).

4. Dimensions of arrays?

- 1D → [1,2,3]
- 2D → matrix
- 3D → cube of numbers

5. Difference between 1D, 2D, 3D?

- 1D: single row
- 2D: rows + columns
- 3D: multiple 2D arrays

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6. How to create array?

Using `np.array()`

7. List vs Array?

- List: slow, mixed types
- Array: fast, same type

8. Properties of array?

shape, size, dtype, ndim

9. dtype?

Data type of elements in array

10. Different data types allowed?

Yes but NumPy converts them to a single type

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11. Element-wise operation?

Operation applied to each element individually

12. Scalar addition?

Adds value to every element

13. Broadcasting?

Automatic expansion of smaller array for operations

14. sum, mean, max, min?

Aggregate functions to compute statistics

15. axis?

Direction of operation:

- axis=0 → columns
- axis=1 → rows

🟡 INTERMEDIATE ANSWERS

16. Indexing?

Accessing elements using position

17. Slicing?

Getting a part of array

18. Indexing vs slicing?

Indexing = single element
Slicing = multiple elements

19. Negative indexing?

Access from end (like -1 last element)

20. 2D access?

arr[row][col]

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21. shape?

Structure of array (rows, columns)

22. reshape?

Change shape without changing data

23. reshape vs resize?

- reshape: returns view
- resize: modifies original

24. flatten?

Convert multi-D array to 1D

25. transpose?

Swap rows and columns

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26. zeros()?

Creates array filled with 0

27. ones()?

Creates array filled with 1

28. eye()?

Identity matrix

29. random module?

Generates random numbers

30. random integers?

np.random.randint()

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31. vectorization?

Perform operations without loops

32. why faster?

Uses optimized C backend

33. aggregation?

Operations like sum, mean, etc.

34. standard deviation?

Measure of data spread

35. variance?

Square of standard deviation

🔵 ADVANCED ANSWERS

36. broadcasting?

Perform operations on different shapes automatically

37. example?

[1,2,3] + 5 → [6,7,8]

38. rules?

Match trailing dimensions or expand size 1

39. why NumPy faster?

Uses C-based optimized operations

40. memory efficiency?

Stores homogeneous data in compact form

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41. dot product?

Matrix multiplication of vectors

42. matrix multiplication?

Row × column multiplication

43. dot vs element-wise?

- dot: matrix multiplication
- element-wise: same position multiplication

44. determinant?

Value used to check matrix invertibility

45. inverse?

Matrix that reverses multiplication

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46. eigenvalues/eigenvectors?

Special values/vectors representing transformations

47. importance in ML?

Used in PCA, dimensionality reduction

48. covariance matrix?

Shows relationship between variables

49. normalization?

Scaling values between 0 and 1

50. standardization?

Mean = 0, Std = 1

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51. view vs copy?

- view: same memory
- copy: new memory

52. memory sharing?

Views share same data

53. stride?

Step size between elements in memory

54. ufuncs?

Universal functions applied element-wise

55. masking?

Filtering data using conditions

🔥 EXPERT ANSWERS

56. importance in AI/ML?

Used for fast matrix operations in models

57. memory optimization?

Stores data in contiguous blocks

58. ndarray structure?

Pointer + shape + stride + data buffer

59. speed?

Vectorized C implementation

60. NumPy vs Pandas?

- NumPy: numerical arrays
- Pandas: tabular data

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61. when use NumPy?

When high-speed numerical computation is needed

62. large data handling?

Uses efficient memory + vectorization

63. broadcasting + vectorization?

Removes loops → faster execution

64. ML usage?

Feature processing, matrix operations

65. real-world use?

Image processing, AI models, data science

🟢 BASIC LEVEL PRACTICAL QUESTIONS

1. Create a NumPy array from a Python list.
2. Create a 1D array with values from 1 to 10.
3. Create a 2D array (3×3) with all zeros.
4. Create a 2D array (4×4) with all ones.
5. Create an identity matrix of size 5×5.
6. Create an array using `arange()` from 1 to 20.
7. Create an array using `linspace()` between 1 and 50 (10 values).
8. Find the shape of a given array.

9. Find the number of elements in an array.
10. Find the data type of array elements.

🟡 INDEXING & SLICING PRACTICAL QUESTIONS

11. Print the first element of an array.
12. Print the last element of an array.
13. Extract elements from index 2 to 6.
14. Reverse a NumPy array.
15. Extract even-indexed elements.
16. Print a specific element from a 2D array.
17. Extract a row from a 2D array.
18. Extract a column from a 2D array.
19. Slice a sub-matrix from a 3×3 matrix.
20. Replace a specific element in an array.

🟡 MATHEMATICAL OPERATIONS PRACTICAL QUESTIONS

21. Add two NumPy arrays.
22. Subtract two arrays.
23. Multiply two arrays element-wise.
24. Divide two arrays element-wise.
25. Add a scalar value to all elements.
26. Multiply all elements by 5.
27. Square all elements of an array.
28. Find remainder of division of array elements.
29. Find power of each element in array.

30. Compare two arrays (greater, less, equal).

🟡 STATISTICS PRACTICAL QUESTIONS

31. Find sum of all elements in array.
32. Find mean of array.
33. Find median of array.
34. Find standard deviation.
35. Find variance.
36. Find minimum value.
37. Find maximum value.
38. Find index of min value.
39. Find index of max value.
40. Sort an array in ascending order.

🟠 RESHAPING & MANIPULATION PRACTICAL QUESTIONS

41. Reshape a 1D array into 2D array.
42. Flatten a 2D array into 1D array.
43. Transpose a matrix.
44. Resize an array to different shape.
45. Stack two arrays horizontally.
46. Stack two arrays vertically.
47. Split an array into multiple parts.
48. Concatenate two arrays.
49. Swap rows in a matrix.

50. Swap columns in a matrix.

🔵 ADVANCED PRACTICAL QUESTIONS

51. Perform matrix multiplication.
52. Find determinant of a matrix.
53. Find inverse of a matrix.
54. Solve a system of linear equations.
55. Find dot product of two vectors.
56. Generate random numbers between 1 and 100.
57. Generate a 5×5 random matrix.
58. Normalize an array (0 to 1 scaling).
59. Filter elements greater than 50.
60. Replace all values > 30 with 0.

🔵 CONDITIONAL FILTERING PRACTICAL QUESTIONS

61. Extract all even numbers from array.
62. Extract all odd numbers from array.
63. Find elements greater than average.
64. Replace negative numbers with 0.
65. Count how many elements are > 10.
66. Extract values between 10 and 50.
67. Remove duplicate values from array.
68. Find unique values in array.
69. Replace all zeros with -1.
70. Extract top 3 maximum values.

🔴 REAL WORLD PRACTICAL QUESTIONS

71. Create a marks array of 10 students and find average.
72. Find highest and lowest marks from dataset.
73. Create monthly sales data and find best month.
74. Create temperature dataset and find hottest day.
75. Calculate total sales using NumPy array.
76. Normalize student marks between 0 and 100.
77. Create random dataset of 100 values and analyze it.
78. Simulate dice rolling using NumPy random.
79. Create image-like matrix (grayscale simulation).
80. Find pass/fail students (marks > 40).

🔴 CHALLENGE LEVEL PRACTICAL QUESTIONS

81. Perform matrix addition using loops and NumPy (compare speed).
82. Create a 1000-element array and compute statistics.
83. Build a mini dataset of employees salary and analyze it.
84. Create a confusion matrix manually using NumPy.
85. Simulate stock price movement using random values.
86. Create a 3D array and perform slicing.
87. Implement normalization without using built-in functions.
88. Find correlation between two arrays.
89. Perform broadcasting on mismatched arrays.
90. Build a mini calculator using NumPy operations.

🔥 EXPERT LEVEL PRACTICAL QUESTIONS

91. Perform eigenvalue and eigenvector calculation.
92. Solve linear regression using NumPy only.
93. Implement gradient descent using arrays.
94. Build matrix-based neural network forward pass.
95. Compress image-like array using NumPy.
96. Generate synthetic dataset for ML model.
97. Perform PCA (Principal Component Analysis basics).
98. Simulate dataset scaling and standardization.
99. Build covariance matrix.
100. Implement feature normalization pipeline.
